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Scribble-supervised medical image segmentation based on dynamically generated pseudo labels via multi-scale superpixels

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ABSTRACT

Nowadays, deep learning-based training increasingly requires adequate pixel-level labeled data. However, in the context of medical image segmentation, accurate pixel-level labeling of lesion edges remains a significant challenge for annotators. Nevertheless, making medical image segmentation with weak annotations is one of the most difficult tasks currently because the weak annotations only cover a small portion of the image and contain little relevant information. To maximize the utility of weak annotations, we propose a novel segmentation method that relies on scribble annotations. By utilizing multi-scale superpixels and deep features from the U-Net, the proposed method iteratively expands and generates pseudo labels with higher accuracy and richer information. This process involves similarity calculations, a dynamic adjustment mechanism, and multi-scale refinement for epoch-wise network training. Thus, as the step-wise expansion of high-confident pseudo labels and the elimination of low-confident ones, the performance of the method can gradually approach some fully-supervised methods. Our method outperforms other weakly annotated segmentation methods on the ACDC and ISIC2018 datasets, as shown by extensive experiments. The results show the segmentation performance of the proposed network is superiorly increased by approximately 1.8%, 3.5% and 1.6% on *IoU*, *CPA* and *Dice*, respectively, and the 95% hausdorff distance (HD95) decreased by approximately 0.8. Furthermore, ablation experiments confirm the effectiveness of each component of our method.

1. Introduction

Medical image segmentation plays a pivotal role in computer-aided diagnosis [1,2]. Early medical image segmentation relied mostly on traditional machine learning methods [3]. Nevertheless, with the emergence of fully-supervised deep learning algorithms [4–8], segmentation precision and efficiency have substantially improved, yet it faces significant challenges due to the complexity of pixel-level annotation.

Researchers have put forward several types of segmentation technologies that do not require full supervision, including unsupervised, semi-supervised, and weakly-supervised methods. These technologies aim to tackle the challenge of needing detailed pixel-level annotations. Unsupervised segmentation does not need any labels. It trains models by learning the similarities and differences between samples. While the need for labeling is reducing, getting enough useful data for medical images is tough due to their unclear boundaries, similar textures, and varying object sizes and shapes [9]. Semi-supervised methods use a little bit of labeled data and a lot of unlabeled data. They train models on both, cutting the cost of making precise annotations and outperforming unsupervised methods. However, they still need some pixel-level labels [10–13]. Weakly-supervised methods use easy-to-make labels like image-level, point, scribble, and box annotations, which reduce the expert skills and time-consuming pixel-level annotations. If weakly supervised-based methods can match the performance of unsupervised and semi-supervised methods, that would be a big step forward [14].

In the past, weakly-supervised techniques were mainly used for natural images. Researchers paired simple labels with algorithms like Graph Cut [15,16] and Random Walks [17]. Recently, with deep learning advances, these methods have become popular in many areas. In medical images, weak annotations are used for tasks like cell nucleus segmentation [18], which study added an unsupervised k-means algorithm to expand point annotations, combining weak and full supervision for great results. Upon this work, the authors simplified the labeling process, introduced incomplete point labeling, and incorporated the idea of target detection to expand the point labeling. They also found that using the Dice loss function during training produces better outcomes compared to other loss functions. However, this method is only suitable for a large number of trivial target segmentation, and the application scenarios are limited.

Besides expanding the weak annotations, many studies have trained networks using pseudo labels derived from weak annotations. These

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methods can be categorized into two types: non-iterative and iterative. In non-iterative pseudo-labeling, pseudo labels are generated before the model training begins and are used consistently throughout the training to improve performance. This method relies heavily on high-quality data and is particularly effective for images with clear boundaries [19]. This is because generating pseudo labels in one step for such images can introduce more incorrect information, ultimately leading to subpar results. Iterative pseudo-labeling methods generate and refine pseudo labels over multiple iterations based on image feature classification. However, if the model's feature extraction capability is inadequate, it may introduce errors. As these inaccuracies accumulate through iterations, they can result in progressively worse performance.

Superpixel-based algorithms are often used for weakly-annotated segmentation. They divide images into regions based on color, brightness, and other features [20]. The SLIC algorithm proposed in 2012, is a popular choice. These SLIC-based methods can make relatively accurate pseudo labels and improve segmentation performance, and they have gained significant popularity in this domain [21]. Meanwhile, they highlight the effectiveness of superpixel-based algorithms in weakly-annotated segmentation tasks, where they facilitate the accurate generation of pseudo labels and contribute to the improvement of segmentation networks.

However, superpixel-based segmentation works well only in specific cases, especially with natural images. It struggles with complex medical images. In these medical images, lesions can cover areas with different textures and varying intensities. As a result, the pseudo labels created may not accurately trace the edges of lesions. This is because the texture is complicated and the edges of lesions are blurry at the single superpixel scale.

To solve the problems mentioned, we introduce a new method for weakly-supervised segmentation using scribble annotations. Specifically, we start by dividing the image into multi-scale superpixels using the SLIC algorithm. Then, we combine these super-pixels with scribble annotations to create initial pseudo labels. Next, we use deep features from a U-Net segmentation network to expand these initial pseudo labels. To handle inaccuracies caused by similar features in different superpixel regions, we have developed a dynamic adjustment mechanism. This mechanism considers two factors: the time factor and global similarity. The time factor adjusts how much we rely on the network's depth features, especially useful as the network learns over time. Global similarity looks at the feature distribution across the entire dataset, preventing reliance on features from uncommon samples that might be off. This adjustment mechanism helps remove pseudo labels that are not confident enough, which boosts accuracy. Finally, we create refined pseudo labels by combining the generated pseudo labels with the network's predictions at different scales. A voting process is introduced to decide the final label for each training step. The multi-scale, dynamic pseudo-labeling approach overcomes the limitations of previous methods. It prevents errors from building up and avoids poor network training that can happen when scribble annotations do not provide enough information.

Our contributions can be summarized as follows:

1. We introduce a novel method to segment medical images using scribble annotations. The proposed method greatly lightens the load of manual annotation and tackles the issues of getting large, detailed medical image datasets.

2. Our method uses superpixels and deep features to generate high-confident class-specific pixels. It iteratively expands and creates more accurate pseudo labels with more details through calculating similarities, adjusting dynamically, and refining at multiple scales. In the dynamic adjustment mechanism, the time factor and global similarity parameters ensure the accuracy of deep features. By removing low-confident pseudo labels, our method can fixe issues with edge prediction and artifacts in other scribble-based methods, especially for targets with similar textures or shapes.

3. We compared our method with other segmentation techniques, both weakly-supervised, semi-supervised and fully-supervised ones. The results on medical image datasets are excellent, nearly matching fully-supervised methods. Additional ablation studies also confirm that our way of making pseudo labels works well.

In the following sections, Section 2 describes the related works on the weakly-supervised segmentation techniques. Section 3 presents the proposed method in detail. Sections 4 and 5 describe the experimental results and discussion. In Section 6, we conclude our method and future works.

2. Related works

2.1. Weakly-supervised segmentation techniques in natural images

Weakly-supervised segmentation began with natural images and has progressed nicely. Regarding the point annotation data, researchers proposed a loss function to measure how close pixels are to point annotations, which helped segment remote sensing images from satellites [22]. Similarly, a team developed a method called Region Importance Sampling (RIS) to segment natural images using box annotations [23]. They used this weakly-supervised model to label parts of the image. Another group created an algorithm for segmenting infrared images of insulators, showing good results [24]. When it comes to scribble annotations, a method was proposed for segmenting aerial images of buildings, using techniques that compete against each other [25]. After that, a comprehensive approach was introduced to reduce uncertainty and segment natural images based on scribbled data [26].

Building on this, an attention module and loss function were created for scribble annotations, allowing networks to train with only scribble annotations [27]. Additionally, the seed diffusion method was used to gradually expand and improve scribble annotations [28]. This idea of expanding annotations as the network learns has become popular. For road datasets, superpixels were used to generate and refine pseudo labels, treating uncertain pixels as unknown [29]. A significant approach involved creating a model like the superpixel method by using scribble-annotated data to pass information to unlabeled pixels [30]. It led to promising results in segmenting the PASCAL-CONTEXT dataset, with scribble annotations showing better versatility in complex targets.

However, medical images pose extra challenges compared to natural images. They often have more complex targets and backgrounds, along with more noise and similar textures. The above segmentation methods for natural images generally exhibit inaccurate boundaries and the presence of artifacts when facing more complex medical images.

2.2. Weakly-supervised segmentation techniques in medical images

For medical image segmentation, researchers trained segmentation networks with improved point labels to better use label data [19]. For heart segmentation, a mix of full and weak annotations was used, with a two-level 'teacher-student' network setup [31]. In meibomian gland segmentation, a two-stage approach was proposed, a simple network to detect gland shape, followed by a complex network for precise segmentation [32]. In 3D image segmentation, incomplete pixel annotations were used to train networks [33]. Scribble-based annotation filtering was introduced to expand scribbles into reliable pseudo labels for training [34]. An attention and gating mechanism with adversarial loss was proposed to boost network performance in medical image segmentation [35]. Transformation consistency was used to develop a lightweight model for segmenting COVID-19 lesions in CT images [36]. Apart from the aforementioned complex network structures, Can et al. [37] discovered that training the network solely with scribble annotations can yield excellent results. These latest studies have optimized segmentation networks from multiple perspectives and demonstrated good results in different application scenarios. However, when dealing

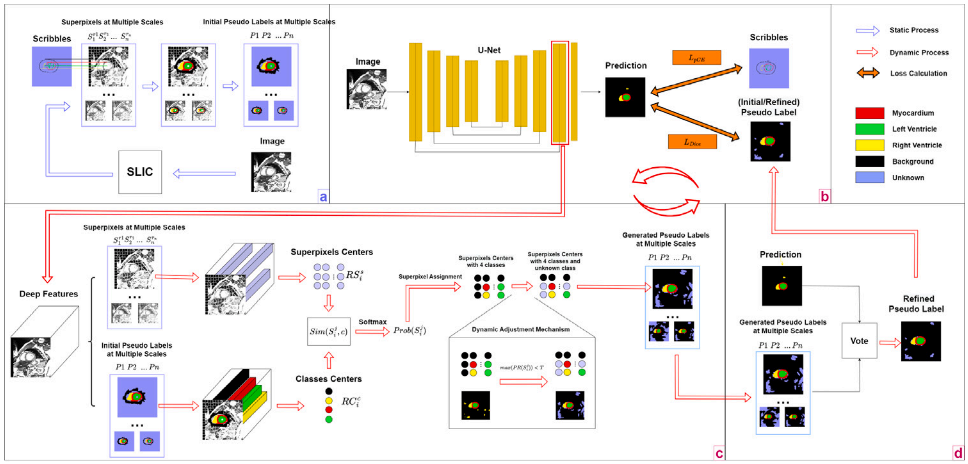


Fig. 1. The illustration of the proposed method on cardiac ACDC dataset. (a) demonstrates the process of initial pseudo label generation, which is obtained with superpixels and scribble annotations. The iterative pseudo-labeling process is shown in (b), (c), and (d). In (b), the deep feature extraction based on U-Net is demonstrated. (c) is the expansion of the generated pseudo labels. (d) illustrates how to refine pseudo labels based on the voting approach at multiple scales. Zoom in to view more clearly.

with multiple types of data, there is often still significant volatility in model performance.

Superpixels have been integrated into segmentation with loss functions to update networks. A loss function using superpixel relationships was designed [30]. Then, the superpixel-based model categorized pixels into superpixels, creating pseudo labels with deep feature information [38]. This thread led to promising results in cell segmentation. In 2022, double decoder structure was proposed to combine deep features from two decoders for scribble annotations [39]. Feature maps were extracted, classified, and upscaled to create new pseudo labels, leveraging multi-scale feature maps for deep information [40]. In 2023, high-precision network, ScribbleVC, combined CNN and Transformer features, integrating scribbles and class embeddings for better accuracy [41]. In 2024, another pre-segmentation strategy was introduced to enhance pseudo labels and network performance with minimal costs [42]. Nevertheless, the accurate acquisition of superpixels would directly affect the training effect, which is still a difficult issue in the superpixel-based weakly supervised method.

3. Methodology

In this section, we present our new segmentation method, which only uses scribble-annotated data. First, the SLIC algorithm is used to divide the image into multi-scale superpixels. And the initial pseudo labels are made by mixing superpixels with scribble annotations and then expand these labels by grouping superpixels that have similar deep features. To remove uncertain pseudo labels, we use a dynamic adjustment mechanism. The final refined pseudo labels are achieved through a voting process at different scales. These refined labels have more information and are more accurate, making them very effective for training the segmentation network. Throughout this process, we use the U-Net architecture for deep feature extraction and segmentation. The flowchart of the proposed method is shown in Fig. 1.

3.1. Initial pseudo label based on superpixels and scribble annotations

The SLIC algorithm [21], depicted in Fig. 1(b), used the CIE Lab color space to partition an image into several super-pixels based on clustering using a pre-set number of seeds. SLIC can adapt to different scales of super-pixel requirements by adjusting parameters, and compared to other super-pixel algorithms, multi-scale partitioning is more flexible.

Here, this algorithm divides the original image into different scale-specific superpixels denoted as S_j^i. S_j^i refers to the jth superpixel at the ith scale. And X is denoted as each pixel in the image. $C_s(x) = c$ indicates the pixel X in the scribble annotation is labeled as class c. To ensure consistency among the superpixel scales, we define a set of pseudo labels as $P_1, P_2, \ldots, P_i, \ldots, P_n$ at n scales. The class of pixel X in each pseudo label is denoted as $C_{P_i}(X)$. That is, $C_{P_i}(X) = c$ signifies that in the pseudo label P_i, the pixel X is labeled as class c at scale i.

Based on the principles of the SLIC algorithm, pixels within the same superpixel are more likely to possess similar semantic information. Leveraging the scribble annotation, classes are assigned to pixels in corresponding superpixels to produce the initial pseudo label. Here, $Unknown$ represents an unknown class that such pixels are not involved in network training.

Specifically, in the scribble annotation, when considering the area corresponding to superpixel S_i^j, only one known class c exists within this area. In the pseudo label P_i, all the pixels within the area corresponding to superpixels S_i^j are labeled as class c, while any other pixels outside that area are marked as $Unknown$. This set of pseudo labels represents the initial pseudo labels. r_i denotes the total amount of superpixels at the ith scale of n scales, where $1 \le i \le n$, $\quad r_1 \cdots j \cdots r_n$.

3.2. Expansion of the generated pseudo labels based on feature center similarity

We employ the U-Net as the backbone network, which consists of an encoder part and a decoder part. The network performs image segmentation by extracting deep feature information from the image. In our method, we focus on extracting feature information from a specific layer of the U-Net decoder. The feature maps are then upscaled to the original image size, where each pixel corresponds to a feature vector denoted as $F(X)$ from the layer of U-Net. To illustrate the process of pseudo label expansion based on clustering and feature similarity, refer to Fig. 1(c).

The class center RC_i^c, as shown in Eq. (1), represents the average feature of class c at ith scale.

$$RC_i^c = \frac{\sum F(X_c)}{|X_c|} \tag{1}$$

In the previous step, superpixels were assigned to initial pseudo labels. At the ith scale, the area corresponding to superpixels S_i^j was divided into the c class or the $Unknown$ class. The feature vector $F(X_s)$ corresponding to all pixels within the superpixel S_i^j is obtained, and the mean of these feature vectors is computed as the center of the superpixels. RS_i^s represents the central feature of the s superpixel at the ith scale, its calculation is described as follows:

$$RS_i^s = \frac{\sum F(X_s)}{|X_s|} \tag{2}$$

The similarity between the class center and the superpixel center is calculated in Eq. (3) and thus obtain the similarity between each superpixel and pixels on each class. $Sim(S_i^j, X_c)$ represents the similarity between superpixel S_i^j and pixels X on class c. The greater the similarity is, the more likely the class of pixels contained in superpixel S_i^j is c.

$$Sim(S_i^j, X_c) = \frac{RC_i^c \cdot RS_i^s}{|RC_i^c| * |RS_i^s|} * \frac{1}{\gamma(S_i^j, X_c)} \tag{3}$$

where $\gamma(S_i^j, X_c)$ represents the distance weight of superpixel S_i^j relative to pixels X on class c at scale i in Eqs. (4)–(5).

In Eq. (4), $D(S_i^j, X_c)$ represents the distance between superpixel S_i^j and pixels X on class c. It is defined as the minimum distance between superpixel S_i^j and pseudo label P_i on class c at the ith scale. If there are no pixels of class c in the pseudo label P_i, the distance is considered infinite. The term $dist$ refers to the Euclidean distance. The term $maxD$ indicates the maximum distance among all superpixels to corresponding class. Similarly, the term $minD$ denotes the minimum distance. Each superpixel's distance relative to each class is normalized by dividing it by $maxD$ and $minD$. The adjustable parameter γ_1 is added to modify the numerical distribution of the distance, ranging from γ_1 to $\gamma_1 + 1$. A higher distance weight indicates that the representative superpixel S_i^j is further from the pixels of class c, resulting in a lower probability of the superpixel belonging to class c. To reduce the similarity of superpixels with large distance weights, the similarity calculated in Eq. (4) is divided by the distance weight.

$$D(S_i^j, X_c) = \begin{cases} minD(dist(S_i^j, X_c)) & \text{, if } \exists C_{P_i}(X) = c \text{ and } \forall X \in S_i^j \\ \infty & \text{, other} \end{cases} \tag{4}$$

$$\gamma(S_i^j, X_c) = \begin{cases} \frac{D(S_i^j, X_c) - minD_c}{maxD_c - minD_c} + \gamma_1, & \text{if } \exists C_{p_i}(X) = c \text{ and } \forall X \in S_i^j \\ \infty & \text{, other} \end{cases} \tag{5}$$

The similarity matrix is further processed and presented in the form of probability through the Softmax function, as shown in Eq. (6), where the probability of superpixel S_i^j relative to each class is calculated by $Prob(S_i^j)$ as follows:

$$Prob\left(S_i^j\right) = \text{Softmax}\left(Sim(S_i^j, X_c)\right) \tag{6}$$

For superpixels S_i^j, the class corresponding to the highest probability in $Prob(S_i^j)$ is most likely to be the real class. The corresponding area in the pseudo label P_i is assigned to same class to realize the expansion of the pseudo label, as shown in Eq. (7).

$$C_{p_i}\left(S_i^j\right) = C_{max}\left(Prob\left(S_i^j\right)\right) \tag{7}$$

where C_{max} represents the class corresponding to the maximum probability.

3.3. Elimination of low-confident pseudo labels by dynamic adjustment mechanism

A dynamic adjustment mechanism can adapt the pseudo-labeled information to eliminate the low-confident pseudo labels, as viewed in Fig. 1(c). The relationship between the maximum probability value $max(Prob(S_i^j))$ and a threshold T is defined through Eq. (8). When this maximum probability value falls below the threshold value, the proposed mechanism recommends dynamically adjusting all corresponding superpixel S_i^j in pseudo label C_{P_i} to the $Unknown$ class. It is noted that the pixels which belong to $Unkonwn$ class are not involved in network training. That step can guarantee the elimination of low-confident pseudo labels.

$$C_{P_i}\left(S_i^j\right) = Unknown, \text{ if } max\left(Prob\left(S_i^j\right)\right) < \text{T} \tag{8}$$

In the proposed method, the threshold T is influenced by two factors, the time factor and the global similarity, to handle the variations in quality of deep features through different epochs. Regarding to the time factor, the expansion of pseudo-labeling is based on the deep features. In the early epoch, due to the poor deep features, the reliability of the superpixel center RS_i^s and the class center RC_i^c are all low, setting a higher threshold can guarantee the areas with the highest probability value. As the training processing, the threshold T is gradually lowered, and the pseudo labels expansion gradually cover larger high-confident areas.

In addition, the global similarity is defined as the global class center RC_G^c to determine low-confident areas from broader perspective, which is shown in Eq. (9).

$$RC_G^c = \frac{1}{|\text{M}|} \sum \text{RC}_i^c \tag{9}$$

where RC_G^c is the global class center of the c class, which is obtained by collecting the average of all class centers RC_i^c of all images in the entire training set. |M| is the number of images in training set.

The global similarity $Sim_G(RC_i^c, RC_G^c)$ represents the similarity between the class center RC_i^c and the global class center RC_G^c on the c class of the current image, as shown in Eq. (10).

$$Sim_G(RC_i^c, RC_G^c) = \frac{RC_i^c \cdot RC_G^c}{\left|RC_i^c\right| * \left|RC_G^c\right|} \tag{10}$$

In order to assess the similarity between classes of current image and the corresponding global class center, it is necessary to measure their similarity. When there is a high degree of similarity, it is indicative of the features of the current image being identical to the majority of the images in the training set, which implies that as the network can learn more reliable features, and the accuracy of feature extraction is enhanced. Conversely, a low degree of similarity between the global class center and the class center demonstrates that the features of the current image diverge significantly from those of the majority of the images. In such cases, it is imperative to increase the probability threshold to reduce the scope of expansion in pseudo-labeling as well as ensure the accuracy of the expansion area.

In light of the aforementioned characteristics, we implement a dynamic threshold adjustment mechanism in Eq. (11). The iteration and max_iterations are employed to denote the current iteration and maximum iteration of the network training, respectively. γ_2 is regarded as an adjustable parameter that manages the initial value of the threshold T. In the final stage of network training, the threshold T is maintained proximal to $\frac{1}{|c|}$. It is noted that the influence of the setting of γ_2 is relatively significant in varying datasets. Based on this dynamic adjustment mechanism, in the early stages of model training, priority will be given to ensuring accurate pseudo labeling of high confidence regions. As the training progresses, the scope of annotations will gradually expand. In the later stages of training, the annotation range of pseudo annotations will cover the entire image area as much as possible.

$$\text{T} = \gamma_2 - \frac{\text{iteration}}{\text{max\_iterations}}\left(\gamma_2 - \frac{1}{|\text{c}|}\right) + \left(1 - Sim_G(RC_i^c, RC_G^c)\right) \tag{11}$$

3.4. Pseudo labels refinement at multiple scales

The performance of pseudo-labeling is subject to variation based on the superpixels employed at specific scale. To minimize this possibility of error, a multi-scale refinement approach is employed. As shown in Fig. 1(d), the finally refined pseudo labels P can be obtained by voting on a set of previously generated pseudo labels and the network predicted result. The class with the highest count is taken as the class of the refined pseudo labels for all pixels.

3.5. Loss functions

The training of the segmentation network is carried out through two loss functions. The network's predicted result is respectively compared with scribble annotation and the refined pseudo labels to calculate Dice loss (L_{Dice}) and partial cross-entropy loss (L_{pCE}). The final loss consists of the L_{Dice} and the L_{pCE}. They are expressed in Eqs. (12)–(14).

$$L_{\text{Dice}} = 1 - \text{DiceCoefficient} \quad , \quad \text{DiceCoefficient} = \frac{2|P \cap y|}{|P| + |y|} \tag{12}$$

where $|P \cap y|$ represents the intersection of the pseudo labels P and the predicted result y, and $|P|$ and $|y|$ represent the number of pixels.

$$L_{\text{pCE}} = -\sum_{c} \sum_{i,j \in \omega_{g,p}} \log y_{i,j}^{c} \tag{13}$$

$$L_{\text{final}} = 0.5 \left(L_{\text{Dice}} + L_{\text{pCE}} \right) \tag{14}$$

where $|c|$ represents the number of classes, g represents the scribble annotation, $\omega_{g,p}$ represents the set of marked pixels in scribble g and refined pseudo labels P, and $y_{i,j}^{c}$ represents the probability that pixel ij is class c.

3.6. Network iteration

In the process of iterative training, our method continually enhances its performance by iteratively generating pseudo labels that have higher accuracy and richer information. The initial iteration involves deep feature extraction based on initial pseudo labels, while subsequent iterations involve three steps to generate a new set of pseudo labels: pseudo labels expansion, elimination of low-confident pseudo labels by dynamic adjustment and final refinement. As training continues, the network's overall performance improves. The deep feature extraction also becomes more accurate, it can approach the quality of real labels and eventually stabilizing. The training of the network is stopped via a predefined iteration limit or a threshold.

4. Experimental results and analysis

4.1. Datasets and experimental settings

In order to evaluate the performance of the proposed method, the ACDC [43] and ISIC2018 [44,45] datasets were utilized. Specifically, the images on ACDC dataset, from the 2017 Automated Cardiac Diagnosis Challenge, involved MRI images containing annotations of the end-diastolic and end-systolic cardiac phases, as well as pixel-by-pixel annotations of the left ventricle (LV), right ventricle (RV), and myocardium (MYO). Additional scribble annotations were provided by Valvano et al. [35]. The training set comprised MRI images from 80 patients, with 10 patients each assigned to the validation and test sets. In total, the training set contained 1524 images, while the validation and test sets comprised 112 and 154 images, respectively. This dataset presents significant challenges for weakly supervised segmentation tasks by subdividing the heart region into multiple detailed categories.

The ISIC 2018 dataset, from the Skin Lesion Analysis Toward Melanoma Detection 2018: A Challenge Hosted by the International Skin Imaging Collaboration (ISIC), which consisted of 2594 training images, 100 validation images, and 1000 test images, and involved binary classification data skin cancer images. And Luo et al. [39] provided a program that can simulate manual generation of scribble annotations, thus the scribble annotations were generated based on their corresponding pixel-level annotations. Only scribble annotations were employed for network training. In this dataset, the cancer target area not only has blurred boundaries, but also varies in size, which poses significant challenges for weakly supervised segmentation tasks.

The samples of images and annotations on two datasets are shown in Fig. 2.

The employed image preprocessing involves scaling the images to the size of 256 × 256, followed by mapping the data distribution to the interval [0,1], and application of normalization through mean–variance processing, normMean = [0.485,0.456,0.406] and normStd = [0.229,0.224,0.225]. The data augmentation operation is carried out on the images, involving random rotation of the image in four directions (0°, 90°, 180°, and 270°) and random flipping of the image in both horizontal and vertical directions. Other data augmentation was ignored to avoid texture distortion that can adversely alter medical images.

Regarding the experimental parameters, a total of 100 epochs are conducted for network training. Their optimizer is set as SGD with a momentum parameter value of 0.9. The initial learning rate for the ISIC2018 dataset was 0.03, whereas that for the ACDC dataset was set to 0.01. The hardware is a NVIDIA 1080ti graphic card (11G Memory) and the program runs on Ubuntu 20.04, PyTorch 1.10, and CUDA 11.3.

As for the comparative experiments, we have chosen six weakly annotated methods as benchmarks: pCE, RLoss [46], S2L [34], WSL4MIS [39], USTM [36], and NIT [47]. Specifically, pCE is a foundational method that uses weakly annotated data to train the network. RLoss, introduced by Tang et al. [46] in 2018, integrates a standard regularizer into the loss function, simplifying the training process. S2L, proposed by Lee and Jeong [34] in 2020, generates pseudo labels by averaging network predictions across iterations. WSL4MIS, from Luo et al. [39] in 2022, uses a two-branch network to create high-quality pseudo labels. USTM, from Liu et al. [36] in 2022, improves training with adaptive consistency transformations. Lastly, NIT, by Yang et al. [47] in 2024, uses geometric transformations to optimize the network with pseudo labels. All the semi-supervised methods used 10% the annotated data and 90% the unannotated data for training.

To further demonstrate the differences between weakly annotated, semi-supervised, and fully supervised medical image segmentation methods, we have included four semi-supervised models and two fully supervised models for comparison (AdvEnt [48], DAN [49], MT [12], and UAMT [13]) and two fully supervised models (U-Net [50] and UG-NET [51]). Due to fundamental differences in methodology, data requirements, and training strategies, their performance comparisons are for reference only and should not be directly contrasted.

4.2. Evaluation metrics

To evaluate the performance of the methods, four metrics are employed, namely Intersection over Union (IoU), Dice Coefficient (Dice), Class Pixel Accuracy (CPA), and 95% Hausdorff Distance (HD95). These indicators are widely used in segmentation tasks and can effectively evaluate and reflect the performance of the model. Through them, we can comprehensively understand the segmentation quality of the model.

The IoU metric measures the ratio between the intersection and union of the predicted and actual area of a class and represents the overlap rate between them. The IoU value ranges from 0 to 1. The closer it is to 1, the more accurate the predicted area is. It is formulated as follows:

$$IoU = \frac{TP}{TP + FN + FP} \tag{15}$$

Similarly, the Dice Coefficient measures the degree of similarity between two sets. A Dice value closer to 1 indicates a higher degree of similarity between the two sets. It is used to evaluate the accuracy of the predicted results and ground truth labels in image segmentation.

$$Dice = \frac{2 \times TP}{2 \times TP + FN + FP} \tag{16}$$

The CPA measures the ratio of correctly predicted pixels to all predicted pixels in a class, reflecting the prediction accuracy for each class. Its value closer to 1 indicates a greater accuracy of prediction. CPA calculates accuracy for each class and averages them, giving a more detailed view of model performance across key classes. It can exclude the background class to focus on targets, reducing background influence and providing a more accurate model evaluation. It is formulated as follows:

$$CPA = \frac{TP}{TP + FP} \tag{17}$$

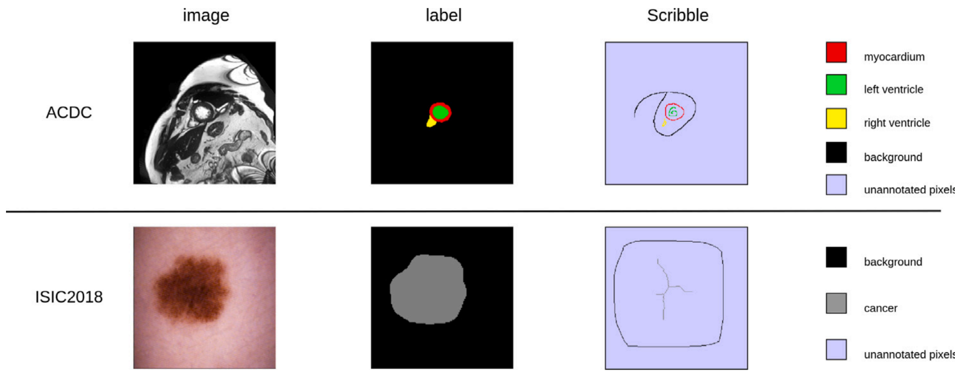


Fig. 2. The samples of the annotations on ACDC and ISIC2018 datasets.

Table 1
The quantitative results on ACDC dataset. Types in the Table: fully-supervised learning (FSL), semi-supervised learning (SSL), weakly-supervised learning (WSL).

Method	Type	Year	IoU	CPA	Dice	HD95
U-Net	FSL	2015	0.754	0.854	0.831	5.91
UG-NET	FSL	2022	0.770	0.875	0.843	4.24
MT	SSL	2017	0.664	0.766	0.747	14.0
DAN	SSL	2018	0.650	0.775	0.734	12.6
UAMT	SSL	2019	0.672	0.779	0.752	13.4
AdvEnt	SSL	2019	0.610	0.728	0.695	14.6
pCE	WSL	–	0.528	0.589	0.657	90.4
RLoss	WSL	2018	0.729	0.817	0.814	6.52
S2L	WSL	2020	0.704	0.789	0.798	18.9
WSL4MIS	WSL	2022	0.704	0.794	0.794	10.1
USTM	WSL	2022	0.535	0.584	0.657	100
NIT	WSL	2024	0.716	0.794	0.804	25.3
OURS	WSL	2024	0.747	0.852	0.830	5.72

Table 2
The quantitative results on ISIC2018 dataset. Types in the Table: fully-supervised learning (FSL), semi-supervised learning (SSL), weakly-supervised learning (WSL).

Method	Type	Year	IoU	CPA	Dice	HD95
U-Net	FSL	2015	0.801	0.888	0.875	20.7
UG-NET	FSL	2022	0.824	0.889	0.892	18.3
MT	SSL	2017	0.708	0.845	0.794	31.0
DAN	SSL	2018	0.732	0.869	0.816	28.6
UAMT	SSL	2019	0.715	0.838	0.799	29.8
AdvEnt	SSL	2019	0.697	0.831	0.786	32.2
pCE	WSL	–	0.732	0.897	0.821	47.5
RLoss	WSL	2018	0.755	0.925	0.835	25.5
S2L	WSL	2020	0.744	0.897	0.832	27.2
WSL4MIS	WSL	2022	0.748	0.887	0.836	24.0
USTM	WSL	2022	0.750	0.882	0.838	28.0
NIT	WSL	2024	0.742	0.917	0.823	25.4
OURS	WSL	2024	0.759	0.911	0.844	22.9

Among them, TP, FP, TN, and FN represent true positive, false positive, true negative, and false negative, respectively.

Finally, the HD95 metric calculates the maximum distance between the predicted and actual areas in the overlapping region, particularly useful for detecting small targets and identifying possible artifacts in the prediction results. A smaller HD95 value indicates a higher overlap between the predicted and actual areas.

4.3. Experimental results

By utilizing the aforementioned experimental setup, we obtained the quantitative and qualitative results of each method. Subsequently, comparative experiments are presented on the ACDC and ISIC2018 datasets in Tables 1 and 2, respectively.

As demonstrated by the Tables 1 and 2, on the ACDC dataset, the proposed method has showcased increases of 1.8%, 1.6%, 3.5% and a decrease of 0.8 in the *IoU*, *Dice*, *CPA* and *HD95* values against the second-place WSL method. And it even approximates the fully-supervised U-Net, which is indicative of significant progress. On the ISIC2018 dataset, our method has demonstrated 0.4%, 0.6% increase in *IoU* and *Dice* values compared with the second-place method, and a remarkable 1.1 drop in the *HD95*.

The results clearly illustrate that the proposed method achieves a superior performance compared to other weakly-supervised and semi-supervised methods. Moreover, the results even indicate that the proposed method approaches the efficacy of fully-supervised methods, specifically in terms of the value of *HD95*, notably highlighting our method in detecting small targets and eliminating artifacts.

The qualitative comparison is depicted in Fig. 3. Our performance is satisfactory on both datasets, despite their considerable differences. This implies that our method can adapt to broad datasets and accomplishing diverse medical image segmentation tasks. In most medical images, accurately distinguishing the boundary of the lesion is a challenge, and differentiating similar targets is another challenge, while some background areas bear similarities with the target's features. Additionally, ISIC2018 exhibits differences characteristics in the shape of cancers. In comparison to other weakly-supervised methods, our method appears to be superior in localizing target positions accurately and removing artifacts consistently. Furthermore, our method can distinguish targets with similar textures with higher accuracy. This is primarily due to the comprehensive information provided by the excellent pseudo labels, which compensates for the lack of scribble annotations, allowing the network to obtain more accurate understanding of the features.

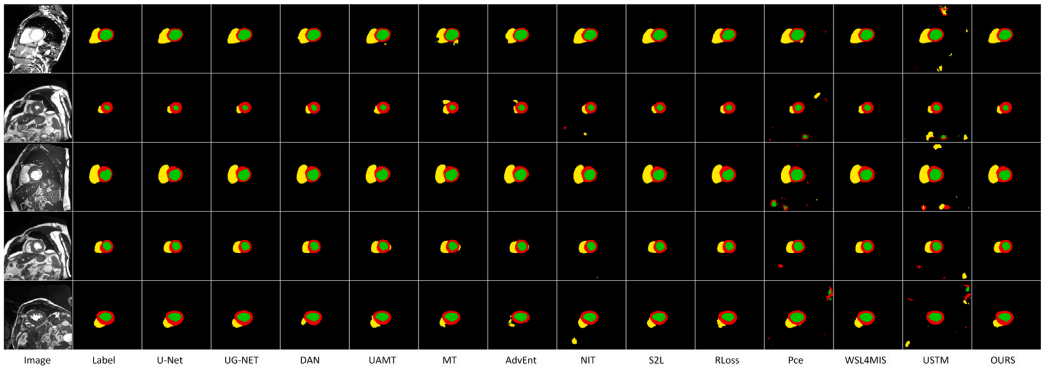
To reveal our method's ability to distinguish the background and the target, the Grad-CAM activation maps are provided to visualize some particular features. The Grad-CAM maps are shown in Fig. 4.

4.4. Ablation experiments

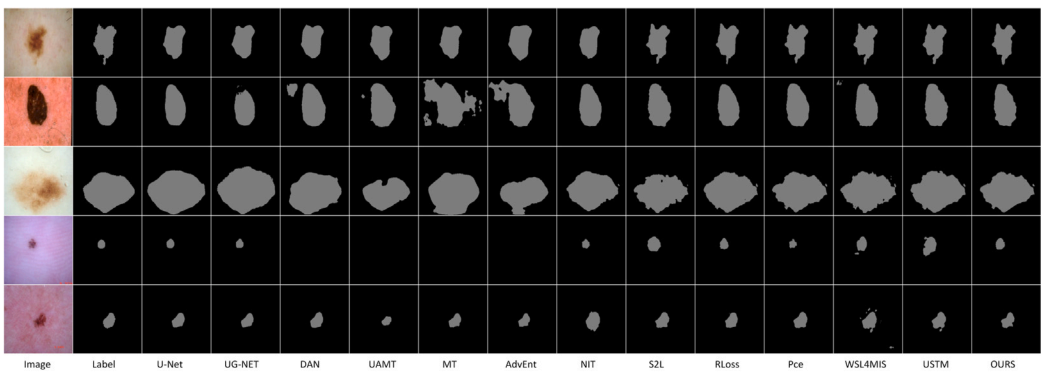
4.4.1. The generation flow of pseudo labels

The generation of a pseudo label can be divided into four stages. Firstly, initial pseudo labels are generated. Secondly, these initial pseudo labels are expanded to create a set of generated pseudo labels. Subsequently, the generated pseudo labels with high confidence and network predicted result are combined and refined to the refined pseudo label for next network training via voting.

The flow of the pseudo-labeling process is depicted in Fig. 5, the images are from two datasets following 100-epoch training. The figure displays the difference in pseudo-labeling at respective stage by using a series of superpixel scales {1024, 1764, 2600, 4096}. The correlation is demonstrated with Fig. 5 in Fig. 1. "A set of initial Pseudo Labels(Phase 1)" corresponds to the initial pseudo-labeling stage in Fig. 1(a), which is based on user-provided scribble annotations. We then showed how "A set of generated Pseudo Labels(Phase 2)" expands upon the initial labels in Fig. 1(c), forming a more detailed set. Finally, "Prediction and

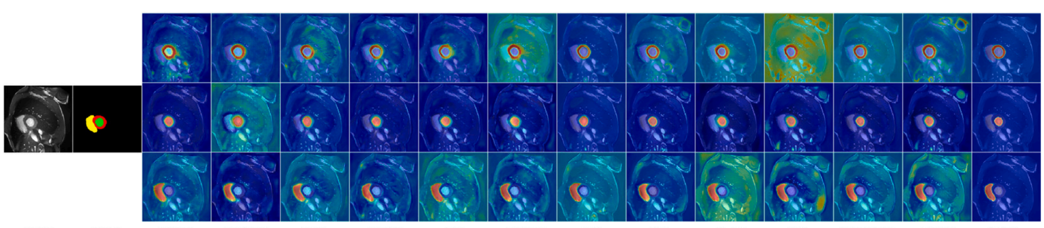


(a) The qualitative results on ACDC Dataset

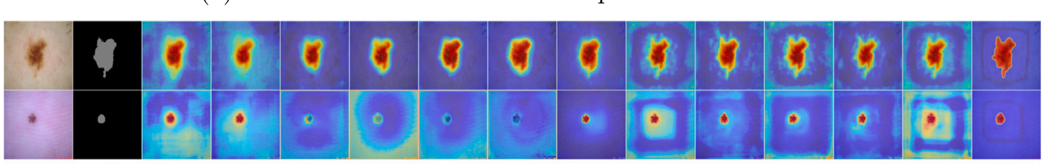


(b) The qualitative results on ISIC2018 Dataset

Fig. 3. The qualitative results. Zoom in to view more clearly.



(a) The Grad-CAM activation maps on ACDC Dataset



(b) The Grad-CAM activation maps on ISIC2018 Dataset

Fig. 4. The Grad-CAM activation maps. Zoom in to view more clearly.

refined pseudo labels" corresponds to the predicted labels and high-quality pseudo labels in Fig. 1(b) and (d), generated through multi-scale fusion strategies for network training.

In addition, the ACDC dataset is utilized to illustrate the proposed dynamic adjustment mechanism, which can lead to a gradual improvement in the accuracy and completeness of pseudo-labeling during network training, as shown in Fig. 6. Specifically, the pseudo labels undergo a gradual expansion during the training process. Initially, they align closely with the original scribble annotations to ensure accuracy in early network training. As the accuracy of features improves progressively during training, the areas with high confidence will enlarge even further. Ultimately, the pseudo labels become the most accurate and comprehensive.

4.4.2. The impact of parameters γ_1 and γ_2 in the network

The setting of the two parameters γ_1 and γ_2 determines the efficacy of pseudo-labeling. And the accuracy of pseudo-labeling directly contributes to the network's segmentation performance. In the experiments, the optimal values of the parameters are influenced by the size and shape of the lesion area in different datasets. To ascertain them, various sets of data were tested. During these tests, we found that the

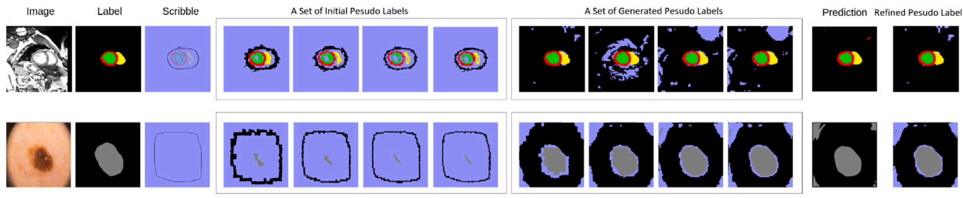


Fig. 5. Generation flow of pseudo labels. Zoom in to view more clearly.

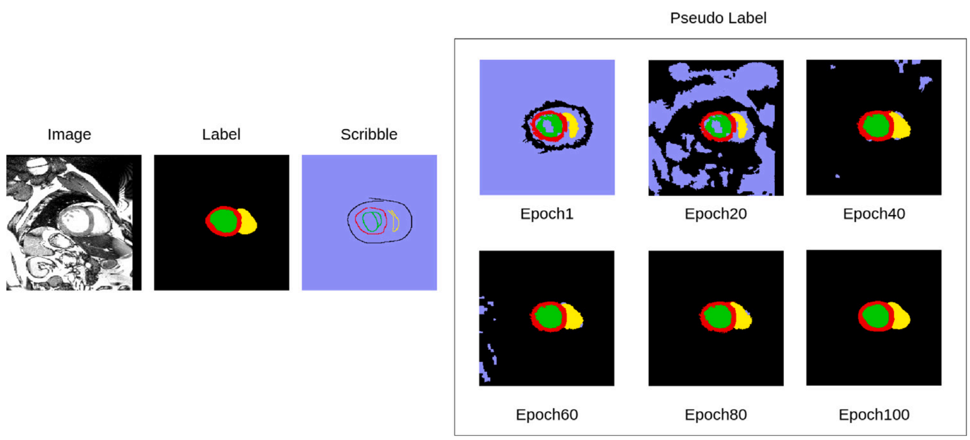


Fig. 6. Pseudo label generation process through epochs. Zoom in to view more clearly.

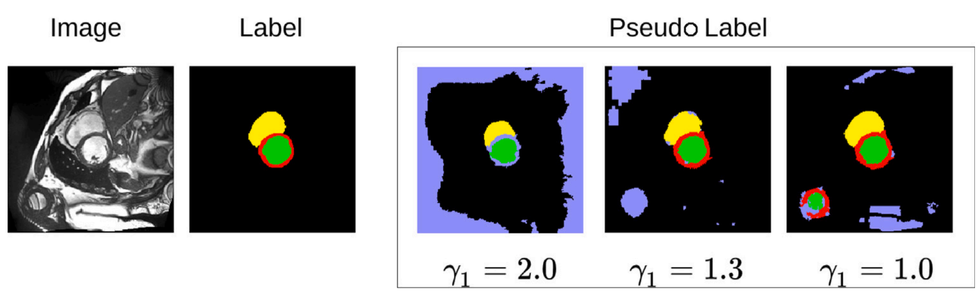


Fig. 7. The illustrations on the impact of γ_1. Zoom in to view more clearly.

values of γ_1 set to 1.3 and γ_2 set to 0.38 yield the best results on the ACDC dataset. Similarly, for the ISIC2018 dataset, the optimal values of γ_1 and γ_2 were determined to be 1.0 and 0.63, respectively.

Substantially, γ_1 is related to the distance weight, and it plays a critical role in the generation of pseudo labels. The impact of varying values of γ_1 is illustrated in Fig. 7. As depicted, a high value of γ_1 results in the dispersion of the distance weights over a vast area. The target superpixels, which are close to the known pixels, have higher probabilities of mislabeling as the background. Conversely, a low value of γ_1 leads to an inadequate impact of distance weights on the superpixels situated further from the known pixels. Some background regions are misclassificated as the target class, particularly on some far background similar to the targets. For example, as shown in Fig. 7, a value of γ_1 set at 2.0 leads to the erroneous labeling of the cardiac part as an unknown pixel due to the high influence of distance weight. A value at 1.0 leads to the emergence of artifacts in the ventricles while a value at 1.3 offers an improved guarantee of the accuracy of both the near and far regions of the targets.

Different values of γ_1 have also been tested on two datasets, and the quantitative results are shown in Tables 3 and 4.

In our method, the parameter γ_2 is responsible for determining the preliminary threshold range and guaranteeing that the final threshold range is approximately equal to $\frac{1}{|c|}$, $|c|$ signifies the number of classes. The impact of γ_2 on the quantity of feature information and precision of pseudo-labeling during training. The 40th epoch on the ACDC dataset is depicted in Fig. 8. When γ_2 is set to a slightly higher value, such as 0.41, the network training is hard to learn novel features. However, a substantial reduction of γ_2 would also lead to erroneous outputs, as evidenced by the presence of artifacts when γ_2 is set to 0.35, which adversely affects subsequent network training as well. A more reasonable parameter setting for γ_2 is 0.38, which ensures the adequate

Table 3
The quantitative impact of γ_1 on ACDC, γ_2 is set to 0.38 simultaneously.

γ_1	IoU	CPA	Dice	HD95
1.0	0.732	0.816	0.819	11.5
1.3	**0.747**	**0.852**	**0.830**	**5.72**
1.5	0.696	0.801	0.782	16.1
2.0	0.661	0.824	0.763	28.5

Table 4
The quantitative impact of γ_1 on ISIC2018, γ_2 is set to 0.63 simultaneously.

γ_1	IoU	CPA	Dice	HD95
0.7	0.739	0.883	0.827	27.0
1	**0.759**	**0.911**	**0.844**	**22.9**
1.3	0.733	0.897	0.821	27.5

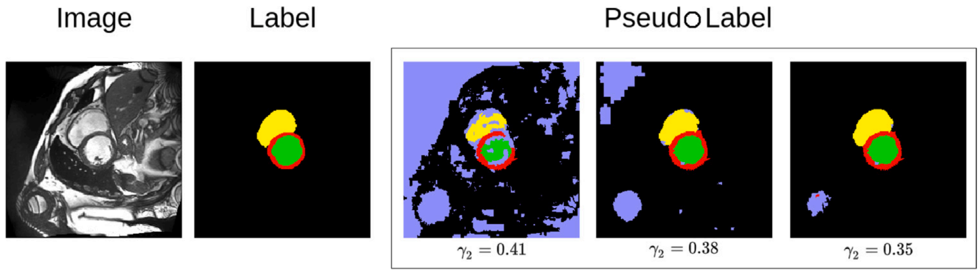


Fig. 8. The illustrations on the impact of γ_2. Zoom in to view more clearly.

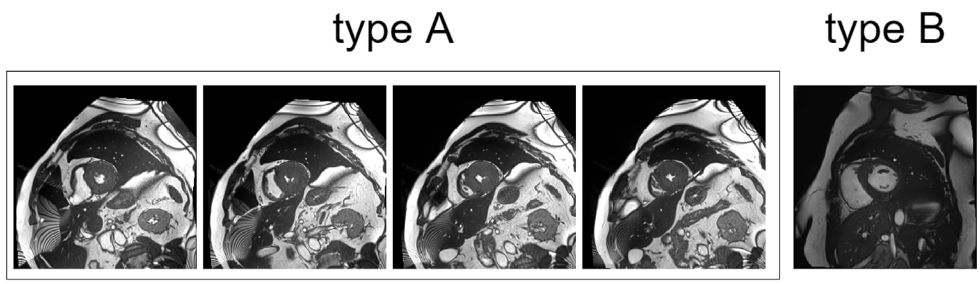


Fig. 9. The examples of two types of images on ACDC dataset: type A (majority) and type B (minority).

Table 5
The quantitative impact of γ_2 on ACDC , γ_1 is set to 1.3 meanwhile.

γ_2	IoU	CPA	Dice	HD95
0.41	0.719	0.819	0.809	7.97
0.38	**0.747**	**0.852**	**0.830**	**5.72**
0.35	0.733	0.819	0.820	8.68

Table 6
The quantitative impact of γ_2 on ISIC2018, γ_1 is set to 1.0 meanwhile.

γ_2	IoU	CPA	Dice	HD95
0.60	0.726	0.875	0.815	33.0
0.63	**0.759**	**0.911**	**0.844**	**22.9**
0.66	0.739	0.877	0.829	34.7

Table 7
The quantitative impact of global similarity on the ACDC dataset.

Global similarity	IoU	CPA	Dice	HD95
√	**0.747**	**0.852**	**0.830**	**5.72**
×	0.642	0.722	0.746	36.8

Table 8
The quantitative impact of global similarity on the ISIC2018 dataset.

Global similarity	IoU	CPA	Dice	HD95
√	**0.759**	**0.911**	**0.844**	**22.9**
×	0.722	0.882	0.814	28.0

amount of feature information and precision of pseudo-labeling. The quantitative results using different γ_2 are presented in Tables 5 and 6.

4.4.3. The impact of global similarity in the network

In clinical practice, it is difficult to achieve optimal pseudo-labeling during training for some unusual images. This is because medical images have more complex and variable features compared to natural images. These features may result in an imbalanced distribution of features in the training set, such as a majority of images Type A and a minority of images Type B in Fig. 9. An imbalanced distribution can significantly disrupt the training process, not only leading to suboptimal training results, but potentially causing the training to fail.

Given the relatively limited number of type B image samples during the training process, and after undergoing the same training cycles, the network's ability to extract their features may be significantly inferior to that of type A images. This can result in the pseudo labels for these images containing more erroneous information. If the same threshold is applied to all images for generating pseudo labels, it may be too low for type B images, thus insufficient to adequately eliminate low-confidence areas. Thus, a certain amount of erroneous information may still exist in the pseudo labels, which is likely to lead to the issue of error accumulation during the training process.

By incorporating global similarity, our method can adjust the pseudo labels generation strategy dynamically, especially for images with minority features like Type B. In other words, global similarity provides additional guidance, helping the network to more accurately identify and reduce errors in pseudo labels. This step enhanced the model's adaptability to various feature images and offers an innovative solution for improving performance in weakly supervised medical image segmentation tasks. We demonstrate the positive impact of the global similarity on the ACDC and ISIC2018 datasets in Tables 7 and 8, respectively. These results indicate a significant improvement compared to lacking it.

5. Discussion

Our experiments demonstrate that our method effectively segments medical images using scribble annotations. The high accuracy of the generated pseudo-labels for complex medical images can be attributed to the combination of multi-scale super-pixels and the dynamic adjustment mechanism integrated into the pseudo-labeling process, as mentioned in Chapter three. This approach not only enhances the precision of the segmentation but also adapts dynamically to the variability present in medical imagery, thereby improving overall performance.

Superpixels at various scales are used to generate initial pseudo labels, each scale offering distinct information. However, with many small superpixels, the labels are more precise but cover less area. Conversely, larger superpixels provide more data but are prone to errors. So there need a trade-off between the accuracy of the pseudo label information and the quantity of information it contains. As the result, when segmentation is conducted via the superpixel algorithm, considerable inconsistency among the superpixels at various scales is inevitable. Our method, based on multi-scale superpixels, considered multiple superpixels from a comprehensive perspective and thus addressed errors instigated at the single scale. Furthermore, the dynamic adjustment mechanism regulates and enhances the pseudo-labeling iteratively, consequently alleviating the cumulative errors during the network training.

For using initial pseudo labels to calculate class centers, as shown in Fig. 1(c), has advantages but also limitations. The goal is to represent the key features of a class accurately through these centers. Directly using scribble annotations might seem suitable for this. However, experiments show that it might not always yield better outcomes than training the network with the initial pseudo labels suggested here. If scribble annotations contain errors or are random, the small number of pixels in them can greatly affect the calculation of class centers. Considering these factors, initial pseudo labels, which include more correct pixels, can minimize the impact of a few inaccurate pixels when calculating category centers. As a result, on many datasets, using these initial pseudo labels for calculating class centers often leads to better segmentation results when superpixel segmentation is suitable.

For the dynamic adjustment mechanism in our method, its first innovation is the flexibility in adjusting the number of pixels covered by pseudo labels at different training stages. Early in training, networks may produce pseudo labelss with many misclassifications due to limited feature extraction capabilities. By dynamically adjusting pseudo labels pixels, we can prevent misclassification accumulation, ensuring more accurate guidance for the network and enhancing training robustness. The second innovation is using global similarity for images with abnormal or unique features. These images, due to their low similarity to others, can still have weak feature extraction capabilities after the same training cycles, increasing misclassifications in pseudo labels. Global similarity helps set higher thresholds for these images, effectively removing low-confidence regions and reducing misclassifications, thus improving model performance on special case images.

During the experimental validation process, we discovered certain constraints within our algorithm. Although the algorithm expands pseudo labels based on the similarity relationship between category centers and superpixel centers, the calculation of category centers relies on the deep features of a set of initial pseudo labels corresponding regions. While using original scribble annotations to define category centers seems more direct and reasonable, experimental results show that using scribble annotations to calculate category centers is less effective than using initial pseudo labels. This is mainly due to two factors: first, scribble annotations may contain inaccurate pixels due to human errors, and since the number of annotated pixels is small, the calculation of category centers using averages is easily affected by individual erroneous pixels; second, scribble annotations are incidental and may not fully cover all feature variations of the lesion area, leading to a deviation of category centers from the actual situation. Therefore, considering the above factors, we chose initial pseudo labels that contain more information and are more robust for most datasets to calculate category centers.

6. Conclusion and future works

In this paper, a new method has been introduced for scribble-supervised medical image segmentation that employs multi-scale superpixels and iteratively generates pseudo labels. This approach enhances the accuracy of pseudo label generation, effectively addressing common issues such as edge prediction errors and artifacts encountered in weakly-supervised segmentation. The method has demonstrated promising results on the ACDC and ISIC2018 datasets, with strong performance observed across various evaluation metrics, thereby providing new insights for clinical applications.

Looking ahead, future research will explore the potential of using image-level annotations for weakly-supervised segmentation. This approach could simplify and streamline the annotation process for medical images, addressing the challenge of limited pixel-level training data. The goal is to enhance the method's applicability and performance in the medical imaging field.

CRediT authorship contribution statement

Zhixun Li: Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Jiancheng Fang:** Writing – original draft, Software, Methodology, Data curation. **Ruiyun Qiu:** Writing – original draft, Visualization, Validation, Formal analysis. **Huiling Gong:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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